

Perform all the Experiments

1. Experiment 1: Introduction to Wireless Network Simulation
 - Overview of NS2/NS3, MATLAB, and OMNeT++ for ad hoc and sensor networks.
 - Understanding network topology setup and performance analysis metrics.
2. Experiment 2: Simulation of Ad Hoc Routing Protocols
 - Implement and analyze AODV, DSR, and DSDV routing protocols.
 - Compare throughput, end-to-end delay, and packet delivery ratio.
3. Experiment 3: MAC Protocol Implementation for Ad Hoc Networks
 - Implement and analyze contention-based MAC protocols (CSMA, MACA, MACAW).
 - Study the impact of collision avoidance techniques in wireless communication.
4. Experiment 4: Energy-Efficient Routing in Wireless Sensor Networks (WSN)
 - Implement LEACH (Low-Energy Adaptive Clustering Hierarchy) protocol.
 - Compare energy consumption and network lifetime.
5. Experiment 5: Localization Techniques in WSN
 - Implement range-based (RSSI, TOA, TDOA) and range-free (DV-Hop, Centroid) localization algorithms.
 - Evaluate accuracy and error rates in node positioning.
6. Experiment 6: Time Synchronization in WSN
 - Implement RBS (Reference Broadcast Synchronization) and TPSN (Timing-sync Protocol for Sensor Networks).
 - Analyze clock drift and synchronization accuracy.
7. Experiment 7: Data Aggregation in WSN
 - Implement data aggregation techniques using clustering methods.
 - Compare efficiency in terms of energy saving and data redundancy.
8. Experiment 8: Security Attacks in Ad Hoc and Sensor Networks
 - Simulate wormhole, blackhole, and Sybil attacks in NS2/NS3.
 - Implement countermeasures for secure routing.
9. Experiment 9: QoS-Based Routing in Ad Hoc Networks
 - Implement QoS-aware routing using bandwidth and delay constraints.
 - Analyze performance trade-offs between QoS metrics.
10. Experiment 10: Performance Evaluation of Sleep-Wake Scheduling Algorithms
 - Implement duty-cycle-based sleep-wake scheduling algorithms.
 - Compare energy efficiency and network lifetime improvement.